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Studierichting: Informatica

Oefeningenreeks 4A nr. 48.4

$$\begin{aligned}\int \ln \left(\frac{x^2}{x^2 - 1} \right) dx &= x \ln \left(\frac{x^2}{x^2 - 1} \right) - \int x d \left(\ln \left(\frac{x^2}{x^2 - 1} \right) \right) \\&= x \ln \left(\frac{x^2}{x^2 - 1} \right) - \int x \cdot \frac{x^2 - 1}{x^2} \cdot \frac{(x^2 - 1) 2x - x^2 \cdot 2x}{(x^2 - 1)^2} dx \\&= x \ln \left(\frac{x^2}{x^2 - 1} \right) - \int \frac{-2}{x^2 - 1} dx = x \ln \left(\frac{x^2}{x^2 - 1} \right) + 2 \int \frac{1}{(x - 1)(x + 1)} dx \\&\rightarrow \frac{1}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1} \\&\Rightarrow A = \frac{1}{1 + 1} = \frac{1}{2} \\&\Rightarrow B = \frac{1}{-1 - 1} = -\frac{1}{2} \\&= x \ln \left(\frac{x^2}{x^2 - 1} \right) + \int \frac{1}{x - 1} dx - \int \frac{1}{x + 1} dx \\&= x \ln \left(\frac{x^2}{x^2 - 1} \right) + \ln |x - 1| - \ln |x + 1| + c\end{aligned}$$

References